

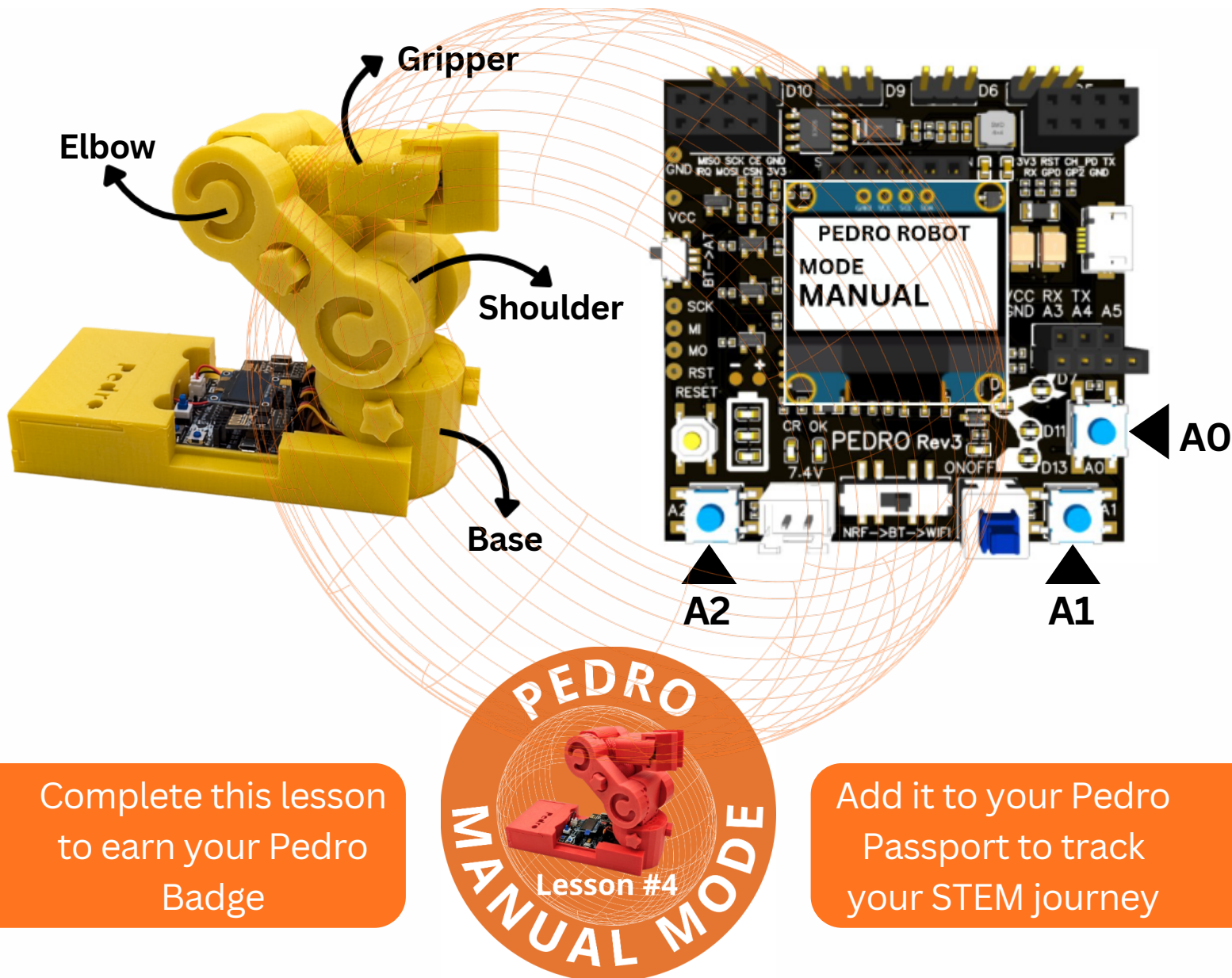
BEGINNER

pedro



STEM Program

Lesson #4: Manual Mode



YOUR MISSION: Learn how to control Pedro robot by using the manual mode.

Learning Goals

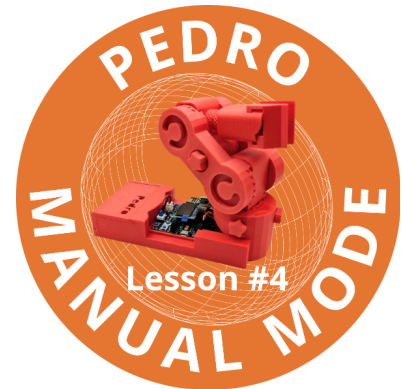
In this lesson, students will assemble the Pedro robot while discovering how robotic arms are designed.

They will explore the mechanical structure of Pedro, learn about servo motors, and understand how planetary gear systems improve motion and precision.

This lesson combines engineering, mechanics, and fabrication in a hands-on experience.

At the end of this lesson, you will unlock your Pedro Badge.

🏆 the Pedro Manual Badge



Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 Assembled** Pedro robot **per student or per group** (*recommended: groups of 2 students*)
 - **OLED Screen 128X64**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



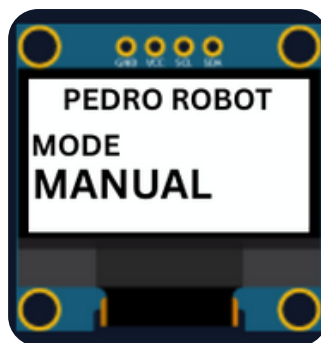
Board Configuration

(Skip this activity if the Pedro program is already uploaded)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions. Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
 - Tools → Manage Libraries → search **PedroRobot** → Install
 - Tools → Manage Libraries → search **U8glib** → Install
 - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✓ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.



Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo. The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

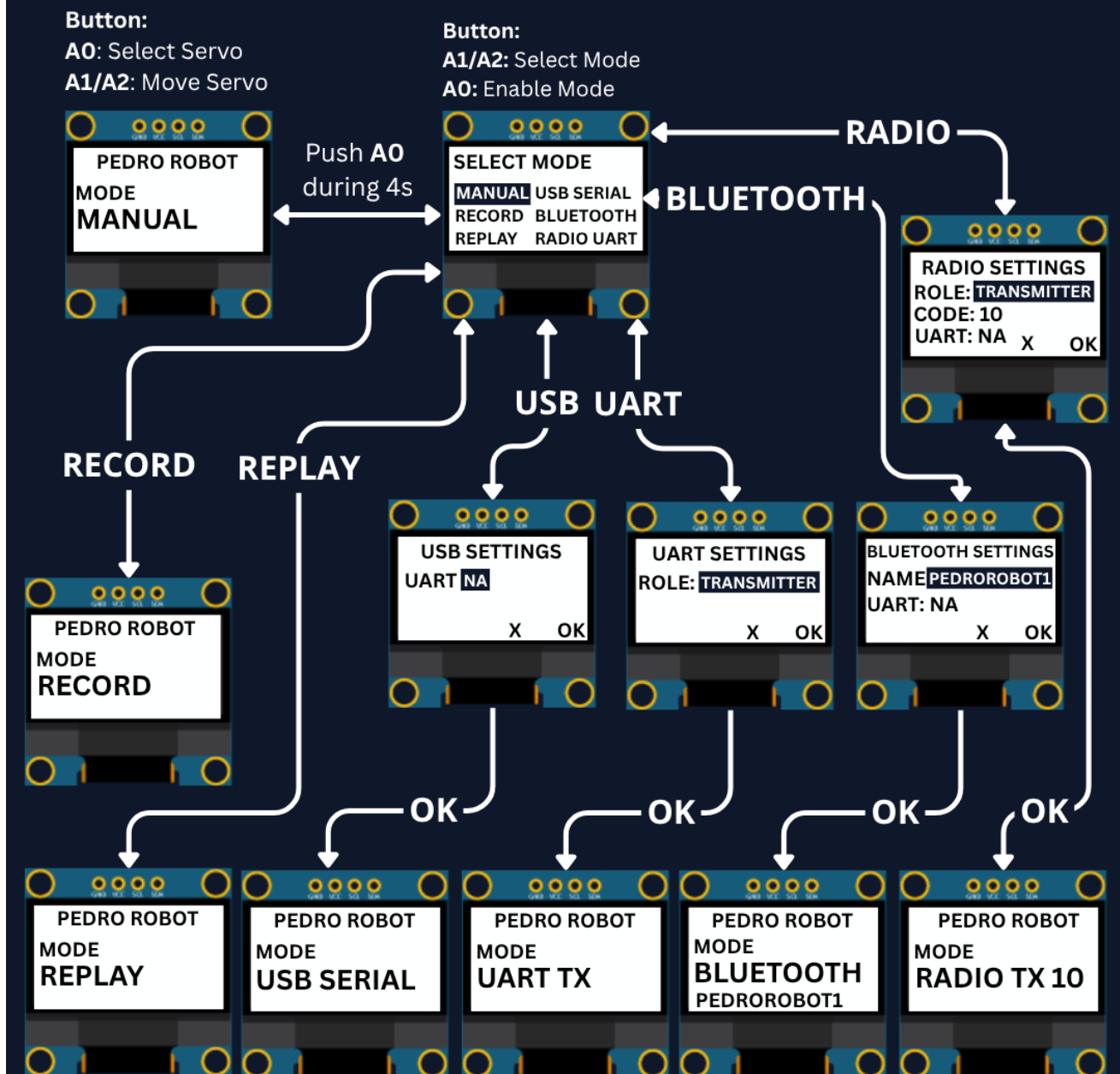
- LED D13 → servo D5 (Base)
- LED D11 → servo D6 (Shoulder)
- LED D8 → servo D9 (Elbow)
- LED D7 → servo D10 (Gripper)

Navigate through the Pedro Robot program. All the program is described in the following illustration. In the next lessons, we will learn how to activate and use the different control modes. For this lesson, we will control Pedro using the **Manual Control** mode.

Pedro Robot

Firmware User Manual

The Pedro firmware allow to control the robot in different modes such as Manual, Bluetooth, USB, UART, Radio and Replay & Record





Control Modes Overview

Using the 128x64 OLED display, users can easily navigate through the menu to select and configure different modes of operation.

The available control modes are:

-  **Radio Mode:** remote control using a radio module **NRF24L01**.
-  **Bluetooth Mode:** remote control via smartphone using module **HC-05**.
-  **USB Serial Mode:** control from a computer using a micro **USB cable**.
-  **Normal Mode:** direct control using onboard buttons.
-  **Record Mode:** record movements.
-  **Replay Mode:** replay movements.
-  **UART Mode:** control by sending data through the **RX** and **TX** pins



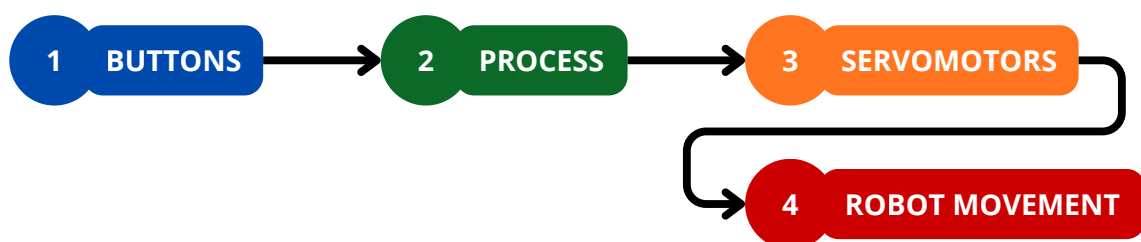
Manual Mode

What it does: Direct control of servo motors using onboard buttons.

How to use:

1. Power ON Pedro (default mode Manual).
2. Press the button A0 to select the servo.
The LED corresponding to the selected servo light on. That button allows you to control each part of the robot. Here is the mapping between the LED ID and the servo ID :
 - LED D13 → servo D5 (Base)
 - LED D11 → servo D6 (Shoulder)
 - LED D8 → servo D9 (Elbow)
 - LED D7 → servo D10 (Gripper)
3. Then use button A1 or A2 to move the selected Servo.

The Manual Mode chain is:

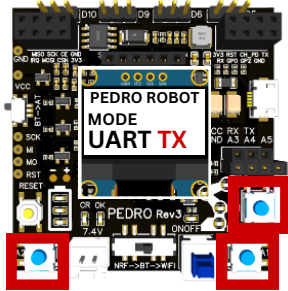




How Manual Mode Works

1

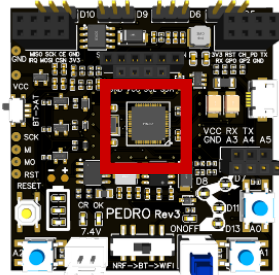
BUTTONS



Control the Pedro Transmitter using buttons **A0**, **A1** and **A2**

2

PROCESS



The **microcontroller** process the data and generates the appropriate **PWM signal**.

PWM Signal

3

SERVO MOTORS



The selected servo receives the **PWM signal** and rotates in the requested direction.

4

ROBOT MOVEMENT

The selected servo rotates, causing the robot to perform the requested movement.





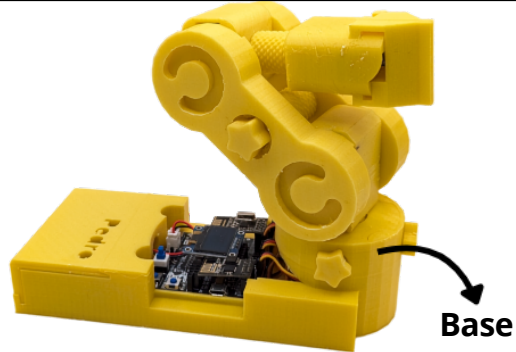
Pedro Action 1 – Move the Base

Student Instructions

- Select the **BASE** servo.
- Move the servo to the **left**.
- Observe Pedro's movement.

What is the INPUT?

The input is the button pressed



Complete the Program

```
if (digitalRead(.....) == .....) {  
  // Button ..... is pressed  
  ACTION → Move servo ..... to the .....  
}
```

What is the OUTPUT?

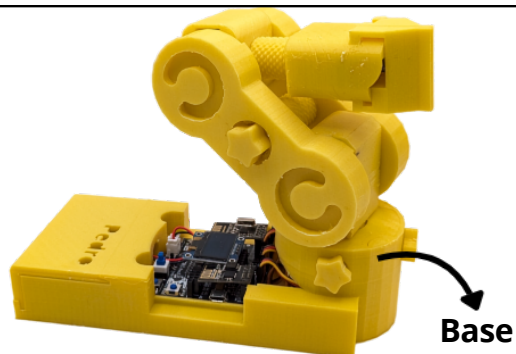
The output is the rotation of the on the

Student Instructions

- Select the **BASE** servo.
- Move the servo to the **right**.
- Observe Pedro's movement.

What is the INPUT?

The input is the button pressed



Complete the Program

```
if (digitalRead(.....) == .....) {  
  // Button ..... is pressed  
  ACTION → Move servo ..... to the .....  
}
```

What is the OUTPUT?

The output is the rotation of the on the



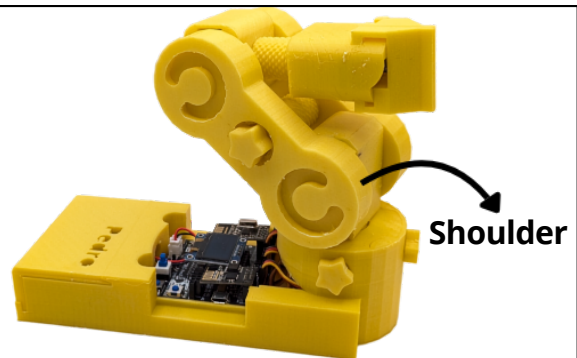
Pedro Action 2 - Move the Shoulder

Student Instructions

- Select the **SHOULDER** servo.
- Move the arm **forward**.
- Observe Pedro's movement.

What is the INPUT?

The input is the button pressed



Complete the Program

```
if (digitalRead(_____) == _____) {  
  // Button _____ is pressed  
  ACTION → .....  
}
```

What is the OUTPUT?

The output is the rotation of the on the

Student Instructions

- Select the **SHOULDER** servo.
- Move the arm **backward**.
- Observe Pedro's movement.

What is the INPUT?

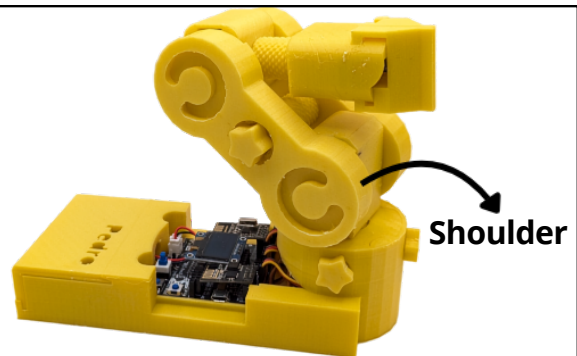
.....
.....

Complete the Program

```
if (digitalRead(_____) == _____) {  
  // Button _____ is pressed  
  ACTION → .....  
}
```

What is the OUTPUT?

.....

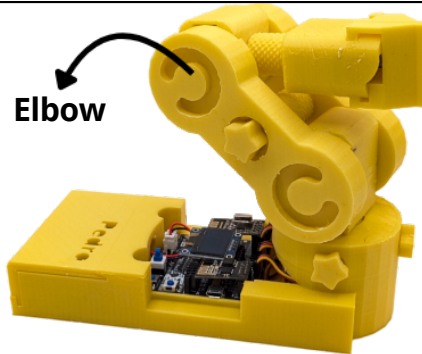




Pedro Action 3 – Move the Elbow

Student Instructions

- Select the **ELBOW** servo.
- Move the arm **up**.
- Observe Pedro's movement.



What is the INPUT?

.....

.....

Complete the Program

```
if (digitalRead(_____) == _____) {  
  // Button _____ is pressed  
  ACTION → .....  
}
```

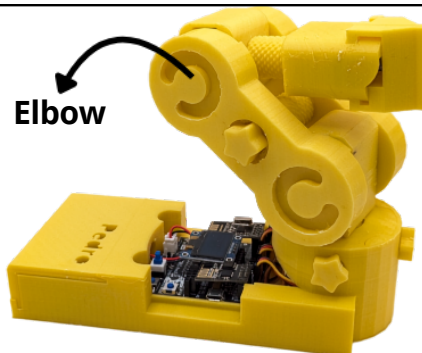
What is the OUTPUT?

.....

.....

Student Instructions

- Select the **ELBOW** servo.
- Move the arm **down**.
- Observe Pedro's movement.



What is the INPUT?

.....

.....

Complete the Program

```
if (digitalRead(_____) == _____) {  
  // Button _____ is pressed  
  ACTION → .....  
}
```

What is the OUTPUT?

.....

.....



Pedro Action 4 – Move the Gripper

Student Instructions

- Select the **GRIPPER** servo.
- **Open** the gripper.
- Observe Pedro's movement.

What is the INPUT?

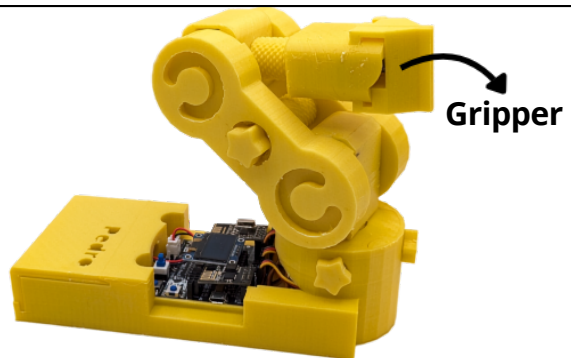
.....
.....

Complete the Program

```
if (digitalRead(_____) == _____) {  
    // Button _____ is pressed  
    ACTION → .....  
}
```

What is the OUTPUT?

.....



Student Instructions

- Select the **GRIPPER** servo.
- **Close** the gripper.
- Observe Pedro's movement.

What is the INPUT?

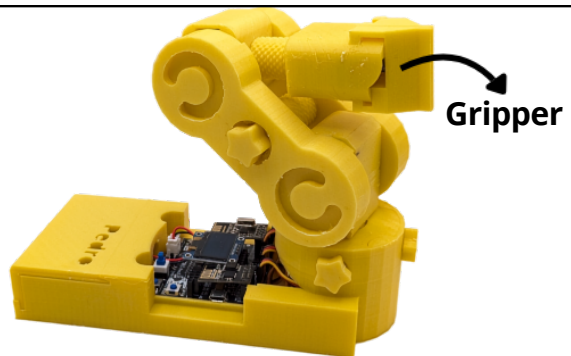
.....
.....

Complete the Program

```
if (digitalRead(_____) == _____) {  
    // Button _____ is pressed  
    ACTION → .....  
}
```

What is the OUTPUT?

.....





Quiz - Check Your Understanding

1. What is the default operating mode when Pedro starts?

- ☐ Bluetooth
- ☐ Radio
- ☐ Manual
- ☐ USB Serial

2. Which button selects the active servo?

- ☐ A0
- ☐ A1
- ☐ A2
- ☐ Reset

3. Which buttons move the selected servo?

- ☐ A0
- ☐ A1 and A2
- ☐ Reset
- ☐ OLED

4. Why does only one LED light at a time?

- ☐ To indicate the selected servo
- ☐ To charge the battery
- ☐ To reset Pedro
- ☐ To increase speed

5. What component acts as Pedro's brain?

- ☐ Servo motor
- ☐ OLED screen
- ☐ Microcontroller
- ☐ Battery

6. Which signal tells a servo where to move?

- ☐ Bluetooth
- ☐ PWM
- ☐ Wi-Fi
- ☐ USB

7. What is the first step after pressing a button?

- ☐ The servo rotates
- ☐ The microcontroller reads the button state
- ☐ The OLED turns off
- ☐ The battery charges

8. Which sequence correctly describes Pedro's operation?

- ☐ Servo → Button → Program
- ☐ Button → Program → Servo
- ☐ Battery → Servo → Button
- ☐ Screen → Program → Battery

9. What does the microcontroller do?

- ☐ Prints the robot
- ☐ Reads inputs and executes the program
- ☐ Powers the battery
- ☐ Moves the gears directly

10. Which real-world machine also follows the Input → Process → Output principle?

- ☐ A 3D printer
- ☐ A robotic arm
- ☐ An automatic door
- ☐ All of the above

Congratulations Mission Completed!

You have unlocked: 🏆 the Pedro Manual Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

